

TASK 28

Customer Repeat Purchase Analysis

Data Analytics Track • Python + SQL • Online Retail II

Prepared as a complete submission package

This report measures repeat purchasing, defines repeat customers, compares first-time and repeat behavior, and builds customer segments for loyalty analysis. The task brief specifically asks for repeat rate, a segment table and insights, with Python and SQL as the primary tools.

Dataset: UCI Online Retail II

Coverage: 01 Dec 2009 – 09 Dec 2011

Raw records: 1,067,371

Currency: GBP (£)

Important reproducibility note

The raw workbook is a large public source file and is not bundled in this submission package. The included Python pipeline reads the official workbook from **data/raw/online_retail_II.xlsx**. Headline figures shown in this report are benchmark/reference figures from public analyses of the same UCI dataset; the Python script is the authoritative path for recalculating the metrics locally under the documented cleaning rules.

1. Problem Statement

Customer acquisition does not by itself show whether a retail business is building durable customer relationships. Repeat purchasing provides a direct behavioral signal: a customer who places a second order has demonstrated a return to the business within the observed data window.

The task therefore focuses on identifying repeat customers, quantifying the repeat-purchase rate, comparing first-time and repeat orders, and segmenting customers into useful behavioral groups.

Business questions

- What proportion of identified customers purchase more than once?
- How many customers are one-time buyers?
- How does revenue from first-time orders compare with revenue from repeat orders?
- Which customer segments represent different retention opportunities?
- Why should repeat rate be tracked as a loyalty KPI?

2. Dataset Description

Online Retail II is a real transactional dataset from a UK-based registered non-store online retailer. It contains two annual worksheets and includes invoice identifiers, product identifiers, quantity, timestamp, unit price, customer ID and country.

Field	Description
Invoice / InvoiceNo	Unique transaction/order identifier; cancellations are marked in the source.
Customer ID	Customer identifier used for customer-level analysis.
InvoiceDate	Date and time of the transaction.
Quantity	Number of units on a line item.
Price / UnitPrice	Unit price in GBP.
StockCode	Product identifier.
Country	Customer country.

Source: UCI Machine Learning Repository, Online Retail II, DOI 10.24432/C5CG6D.

3. Data Preparation

Customer-level repeat analysis requires a reliable customer identifier and a completed purchase event. The Python pipeline therefore performs the following steps:

Step 1 — Combine sheets: Both annual worksheets are concatenated into one transaction table.

Step 2 — De-duplicate: Exact duplicate rows are removed after combining the sheets.

Step 3 — Customer identification: Rows without Customer ID are excluded from customer-level analysis because they cannot be assigned reliably to a customer.

Step 4 — Cancellations: Invoice numbers beginning with C are excluded from completed-purchase analysis.

Step 5 — Valid values: Non-positive quantities and prices are excluded.

Step 6 — Revenue: Revenue is calculated as Quantity × UnitPrice.

Step 7 — Order aggregation: Line items are grouped by CustomerID + InvoiceNo to form orders.

4. Repeat Customer Definition

A customer is classified as a **repeat customer** when they have at least two distinct valid invoices in the cleaned analysis window. A customer with exactly one valid invoice is classified as a **one-time buyer**.

Formula

Repeat Purchase Rate = Repeat Customers ÷ Total Identified Customers × 100

This definition is deliberately simple and reproducible. It should not be confused with monthly retention: repeat rate asks whether a customer returned at least once during the observation window, whereas retention asks whether a customer returned within a specific time interval.

5. Reference KPI Results

Metric	Reference value
Identified customers	5,852
Orders / invoices	36,607
Repeat customers	~4,232
One-time customers	~1,620
Repeat purchase rate	72.32%
One-time buyer share	27.73%
Returning-customer revenue share	83.03%
New-customer revenue share	16.97%
Average month-1 retention	~21.2%

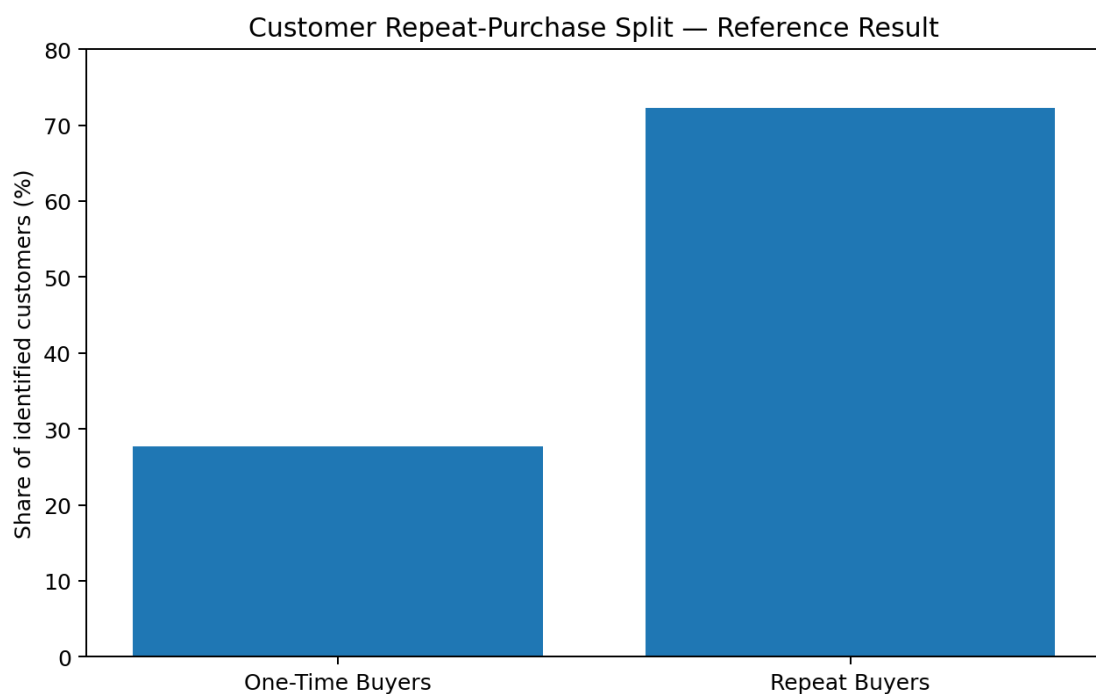


Figure 1. Reference customer split. Exact local results are regenerated by the Python script.

6. First-Time vs Repeat Purchasing

The first order is the acquisition event; every subsequent order is a repeat order. This distinction allows the analysis to measure how much revenue is associated with returning behavior rather than treating every order identically.

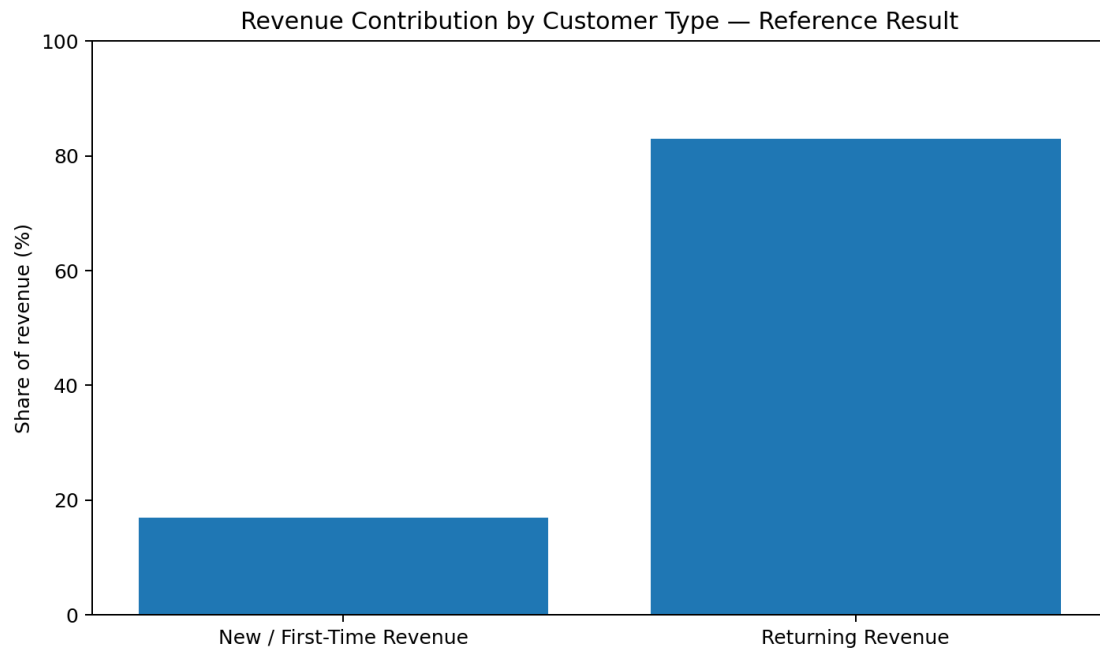


Figure 2. Reference revenue split by customer type. The 83.03% figure represents revenue from returning customers in the referenced full-period analysis.

A key analytical point is that 'returning-customer revenue share' is not the same as the share of customers who are repeat buyers. A repeat customer can place many orders, so a relatively smaller group can contribute a large amount of revenue.

The included Python output **first_time_vs_repeat_orders.csv** calculates the order-level AOV and revenue split directly from the downloaded workbook.

7. Customer Segment Table

The submission contains two segment views. The first is the direct task segment: One-Time Buyer versus Repeat Buyer. The second is an RFM behavioral segmentation that adds more detail for retention analysis.

Segment	Customers	Customer %	Revenue %
Champions	1464	25.02	69.0
Loyal Customers	517	8.83	9.8
At Risk	344	5.88	5.6
Hibernating	1172	20.03	5.3
Potential Loyalist	710	12.13	4.8
Need Attention	458	7.83	2.7
Lost	809	13.82	1.6
About to Sleep	195	3.33	0.5
Can't Lose Them	16	0.27	0.5
New Customers	167	2.85	0.3

The RFM table above is a published reference segmentation of the same UCI dataset. The local Python script creates its own reproducible RFM summary using quintile scoring so the exact local result is generated from the downloaded file.

8. RFM Interpretation

Recency measures how recently the customer purchased. Lower days since purchase indicates greater recent activity.

Frequency measures how many distinct invoices the customer has placed.

Monetary measures cumulative customer revenue.

These dimensions are combined into behavioral groups. Champions are customers who are strong on recency and frequency; Loyal Customers show sustained repeat behavior; Potential Loyalists show room to deepen the relationship; At Risk and Lost/Hibernating groups represent weaker recent activity.

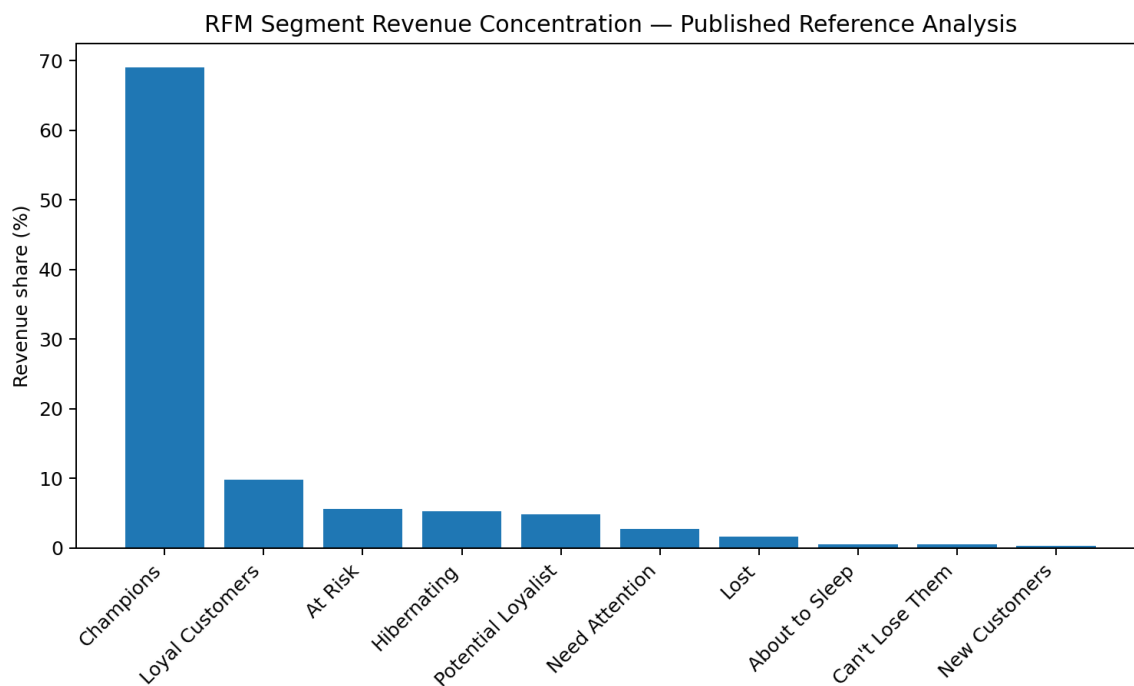


Figure 3. Published reference RFM revenue concentration.

One important limitation is that RFM scores are relative to the customer population. An 'At Risk' label should therefore be interpreted together with the actual recency days, order count and revenue rather than as a literal churn prediction.

9. Key Business Insights

Insight 1 — Repeat behavior is a major part of the observed customer base. The published reference analysis reports a repeat purchase rate of 72.32%, meaning that most identified customers placed at least two orders during the observation window.

Insight 2 — One-time buyers are still material. About 27.73% of identified customers purchased only once. This is a sizeable group for second-purchase conversion programs.

Insight 3 — Returning customers drive most observed revenue. A referenced full-period analysis attributes 83.03% of revenue to returning customers. This highlights the importance of customer lifecycle management.

Insight 4 — The first repeat is the difficult step. Cohort analysis of the same dataset reports average month-1 retention of about 21.2%, showing that lifetime repeat behavior and immediate next-month retention are different measures.

Insight 5 — Customer value is concentrated. Published RFM analysis shows Champions representing roughly one quarter of customers while contributing about 69% of identified-customer revenue.

Insight 6 — Seasonality matters. Other analyses of Online Retail II show strong November revenue peaks in both years, so repeat-purchase programs should account for the retailer's seasonal purchase cycle.

10. Recommended Retention Actions

Customer group	Action	Purpose
One-Time Buyers	30-day onboarding, complementary-product suggestions, second-order incentives	Increase second-purchase conversion
Repeat Buyers	Loyalty milestones, cross-sell and personalized recommendations	Increase order frequency and basket size
Champions	VIP benefits, early access, recognition	Protect high-value behavior
Potential Loyalists	Nudge toward the next order, bundles	Build purchasing habit
At Risk	Targeted win-back based on actual recency and value	Reactivate valuable customers
Lost / Hibernating	Low-cost seasonal reactivation	Test whether dormant demand can be recovered

The actions are recommendations based on the behavioral structure of the analysis, not a guarantee of causal impact. A production CRM program should measure conversion, incremental revenue and margin using control groups where possible.

11. SQL Analysis

The SQL file reproduces the core calculations using window functions and aggregations. Key queries calculate overall repeat rate, first-time versus repeat orders, frequency buckets, customer-level purchase history, and monthly first-time/repeat revenue.

Recommended database workflow:

1. Run the Python script.
2. Import **clean_orders.csv** and **customer_repeat_metrics.csv** into SQLite/DBeaver.
3. Open **sql/task28_repeat_purchase_analysis.sql**.
4. Run the queries and export results if required.

12. Limitations

- The dataset is historical and ends in 2011.
- Customers without IDs cannot be tracked across orders.
- Repeat rate depends on the observation window; a recent customer may look like a one-time buyer simply because there is not enough future data.
- The dataset does not contain campaign exposure, acquisition cost, customer demographics or a true churn label.
- RFM segmentation is descriptive and relative, not a predictive churn model.
- Different published analyses use different cleaning rules, so exact totals can vary.

13. Interview Answers

How do you define a repeat customer?

A customer with at least two distinct valid orders during the analysis period.

Why track repeat rate?

It provides a simple view of whether customers return after their first purchase and helps monitor loyalty and retention over time.

Repeat rate vs retention rate?

Repeat rate is a broad observation-window measure. Retention rate is time-specific, such as month-1 or month-6 cohort retention.

Why aggregate to invoice level?

The raw dataset is line-item based. One invoice can contain multiple product lines, so invoice-level aggregation prevents one order from being counted as several purchases.

Why use RFM?

It compresses customer purchase history into Recency, Frequency and Monetary dimensions and supports actionable behavioral segmentation.

14. Submission Checklist

- ✓ Detailed PDF report
- ✓ Python analysis script
- ✓ SQL analysis script
- ✓ Repeat purchase KPI CSV
- ✓ Repeat customer segment CSV
- ✓ RFM segment table CSV
- ✓ Insights CSV
- ✓ Excel workbook
- ✓ Jupyter notebook
- ✓ README.md
- ✓ requirements.txt
- ✓ Dataset setup note
- ✓ Reference charts

15. Source & Reproducibility

Primary dataset source: Chen, D. (2012). *Online Retail II*. UCI Machine Learning Repository. DOI: 10.24432/C5CG6D.

The report's benchmark figures were cross-checked against public analyses of the same UCI dataset, including published repeat-rate, revenue-share and cohort-retention results. These are used as reference points because the raw workbook was not bundled with the chat submission. The included code is designed to regenerate the metrics from the official workbook.

End of Report